The Journal of Open Source Software

QhX: A Python package for periodicity detection in red noise

Andjelka B. Kovačević[1*¶], **Dragana Ilić**[1], **Momčilo Tošić**[1*], **Marina Pavlović**[2], **Aman Raju**[1], **Mladen Nikolić**[1], **Saša Simić**[3], **Iva Čvorović Hajdinjak**[1], **and Luka Č. Popović**[4]

1 University of Belgrade-Faculty of Mathematics, Studentski trg 16, Belgrade, Serbia **2** Mathematical Institute of Serbian Academy of Science and Arts, Serbia **3** Faculty of sciences, University of Kragujevac, Radoja Domanovića 12, Serbia **4** Astronomical Observatory, Belgrade, Serbia ¶ Corresponding author * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Ivelina Momcheva

Reviewers:

- @danhey
- @sgeorge91

Submitted: 18 November 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

QhX is a Python package for detecting periodicity in red noise time series, developed as an in-kind contribution to the Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST, Ivezić et al., 2019). Traditional Fourier-based methods often struggle with red noise, which is common in quasar light curves and other accreting objects. QhX addresses these challenges with its core 2D Hybrid method (Kovačević et al., 2018). Input data are mapped into a time-period plane via wavelet transforms, which are (auto)correlated to produce a correlation density map in a "period-period" plane. Statistical vetting incorporates significance, upper and lower error bounds, and the novel Intersection over Union (IoU) metric to evaluate the proximity and overlap of detected periods across bands and objects. In addition to a vetted numerical catalog, QhX dynamically visualizes periodicity across photometric bands and objects.

The Journal of Open Source Software

Statement of need

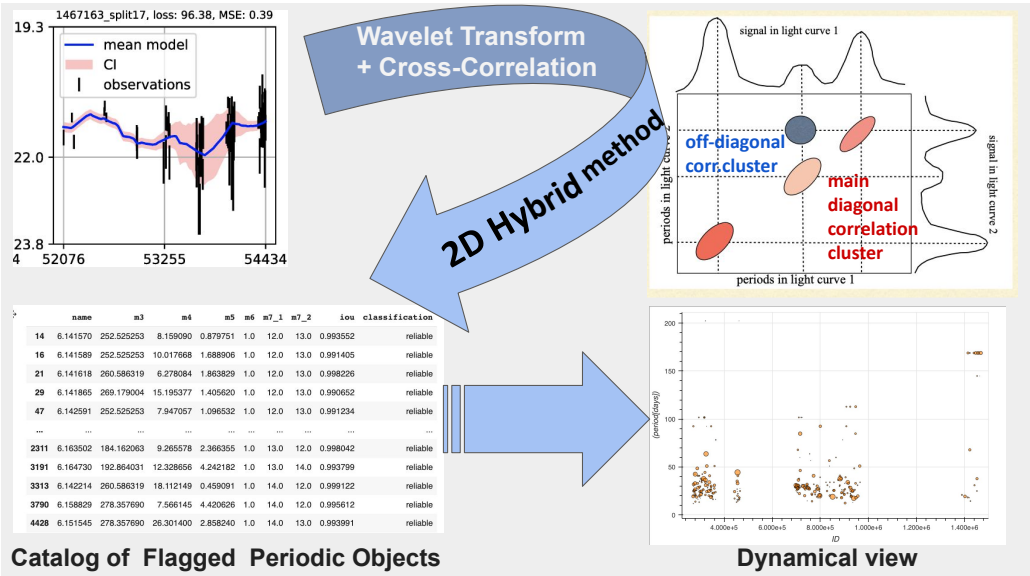

Figure 1: The left panel shows a 1D light curve with observational data (black error bars) and a model (blue line). QhX transforms the time-series into the time-frequency domain and cross-correlates wavelet matrices to produce a 2D period-period correlation map (right), where clusters indicate periodic signals. After map integration, statistical vetting generates a numerical catalog of flagged periodic objects (bottom left) and a dynamic view of detected periods across objects and bands (bottom right).

Periodic variability spans a range of astronomical objects, from asteroids to quasars. Identifying meaningful signals is complicated by red noise (see, e.g., Figure 1 in Gaia Collaboration et al., 2019; Kasliwal et al., 2015; Kovačević, Radović, et al., 2022), which exhibits fractal-like patterns across time scales (Belete et al., 2018; Vio et al., 1991). Non-stationary signals and unfavorable sampling (Brandt et al., 2018; D'Orazio & Charisi, 2023) further obscure coherent patterns. Traditional time-frequency methods, constrained by the Fourier uncertainty principle (i.e., Gabor limit, Gabor, 1947), often fail with such complex signals, highlighting the need for nonlinear approaches (Abry et al., 1995; Cohen, 1995).

QhX provides features specifically designed to address these challenges. The first feature is its core 2D Hybrid method (see Figure 1), detailed in (Kovačević et al., 2018), inspired by 2D Correlation Spectroscopy (Kovačević, 2024; Noda, 2018). By applying wavelet transforms, QhX maps time-series data into the time-frequency domain and (auto)correlates it, generating a period-period correlation density that enhances signal detection. Secondly, QhX introduces an Intersection over Union (IoU) metric, combined with standard statistical measures (significance, upper and lower error bounds), to evaluate the overlap of detected periods across bands and objects. Each period is represented as the center of an "IoU ball," with its radius reflecting relative error, calculated as the mean of the upper and lower error bounds—analogous to a circular aperture in photometry (Saxena et al., 2024). Thirdly, QhX enhances traditional analysis by generating numerical and interactive visual catalogs that rank periodicity candidates by reliability. These interactive catalogs enable detailed inspection of signal consistency, offering greater interpretability than traditional static plots.

DRAFT

QhX structure

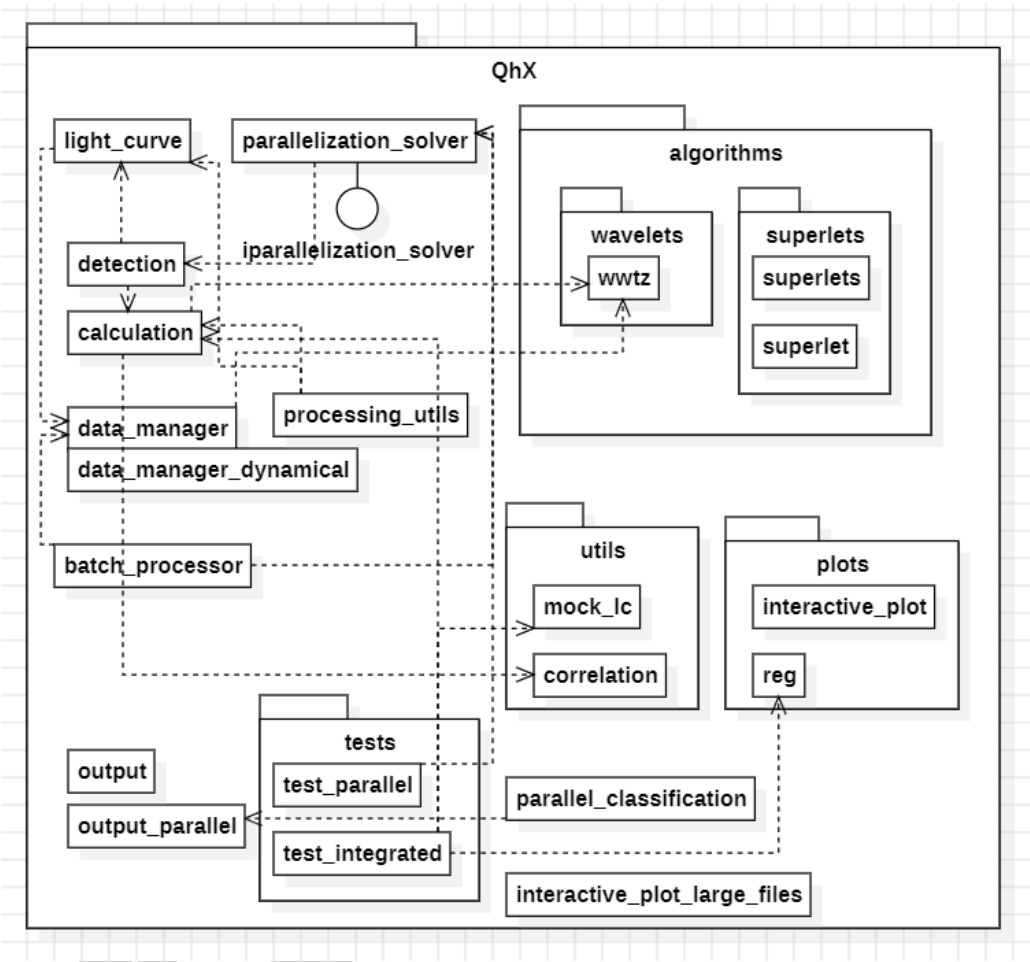

Figure 2: Schematic representation of the QhX package architecture.

`QhX` (Version 0.2.0) is an open-source package optimized for gappy quasar light curves but adaptable to other datasets. It supports both dynamic and fixed modes, with parallel processing capabilities for large-scale data. The modular design facilitates rapid experimentation by enabling easy swapping or modification of functions (see Figure 2), addressing diverse research needs. For fixed-only workflows, specialized functions such as `data_manager` offer minimal overhead and optimal performance, while `data_manager_dynamical` supports both dynamic and fixed configurations to handle more complex scenarios involving dynamic filters.

The package is organized as follows:

1. **Core**:
   - `algorithms` module provides essential signal-processing techniques, including the Weighted Wavelet Z-Transform (`wwtz`) and prototype superlet transforms.
   - `correlation` function within the `utils` module supports the 2D Hybrid method by converting light curve data into wavelet matrices and performing (auto)correlation, creating correlation density.
2. **Signal Detection and Validation**:
   - `detection` module identifies candidate periodic signals and assesses their validity using statistical measures (significance and upper and lower error (Johnson et al.,

2019)). The Intersection over Union (IoU) metric identifies overlapping periods across bands and objects.
    - Statistical vetting categorizes detected periods for each object and band as reliable, medium, or poor.
3. **Data Management**:
    - `data_manager` and `data_manager_dynamical` modules manage data flow, data loading, outlier removal, and format compatibility.
    - `batch_processor` and `parallelization_solver` modules optimize task distribution across multiple processors.
4. **Visualization and Output**:
    - `plots` module includes tools for creating interactive visualizations, such as `interactive_plot`, which allows for exploring detected periodicities across bands and objects. For large datasets, `interactive_plot_large_files` enables in-depth inspection of signal consistency.
    - `output` and `output_parallel` modules handle result storage, supporting both single-threaded and parallelized workflows.
5. **Testing**:
    - `tests` module, containing `test_parallel` and `test_integrated`, validates the functionality across various processing setups.

Representative Applications

The `QhX` method has been applied to:

-Quasar periodicity detection (Fatović et al., 2023; Kovačević et al., 2018, 2019; Kovačević, Popović, et al., 2020). -Quasi-Periodic Oscillations detection (Kovačević, Yi, et al., 2020). -Very Low-Frequency (VLF) signals variability in the vicinity of earthquakes (Kovačević, Nina, et al., 2022). -`QhX` is the LSST directable software in-kind contribution.

Acknowledgements

Funding was provided by the University of Belgrade - Faculty of Mathematics (the contract 451-03-66/2024-03/200104), Faculty of Sciences University of Kragujevac (451-03-65/2024-03/200122), and Astronomical Observatory Belgrade (contract 451-03-66/2024- 03/200002), through grants by the Ministry of Education, Science, and Technological Development of the Republic of Serbia.

References

Abry, P., Gonçalvès, P., & Flandrin, P. (1995). *Wavelets, spectrum analysis and 1/f processes* (A. Antoniadis & G. Oppenheim, Eds.; pp. 15–29). Springer New York. https://doi.org/10.1007/978-1-4612-2544-7_2

Belete, A. B., Bravo, J. P., Canto Martins, B. L., Leão, I. C., De Araujo, J. M., & De Medeiros, J. R. (2018). Multifractality signatures in quasars time series - I. 3C 273. *478*(3), 3976–3986. https://doi.org/10.1093/mnras/sty1316

Brandt, W. N., Ni, Q., Yang, G., Anderson, S. F., Assef, R. J., Barth, A. J., Bauer, F. E., Bongiorno, A., Chen, C.-T., De Cicco, D., Gezari, S., Grier, C. J., Hall, P. B., Hoenig, S. F., Lacy, M., Li, J., Luo, B., Paolillo, M., Peterson, B. M., … Yu, Z. (2018). Active Galaxy Science in the LSST Deep-Drilling Fields: Footprints, Cadence Requirements, and Total-Depth Requirements. *arXiv e-Prints*, arXiv:1811.06542. https://doi.org/10.48550/arXiv.1811.06542

Cohen, L. (1995). *Time-frequency analysis*. Prentice Hall PTR. ISBN: 9780135945322

DRAFT

D'Orazio, D. J., & Charisi, M. (2023). Observational Signatures of Supermassive Black Hole Binaries. *arXiv e-Prints*, arXiv:2310.16896. https://doi.org/10.48550/arXiv.2310.16896

Fatović, M., Palaversa, L., Tisanić, K., Thanjavur, K., Ivezić, Ž., Kovačević, A. B., Ilić, D., & Č. Popović, L. (2023). Detecting Long-period Variability in the SDSS Stripe 82 Standards Catalog. *165*(4), 138. https://doi.org/10.3847/1538-3881/acb596

Gabor, D. (1947). Acoustical quanta and the theory of hearing. *Nature*, *159*(4044), 591–594. https://doi.org/10.1038/159591a0

Gaia Collaboration, Eyer, L., Rimoldini, L., Audard, M., Anderson, R. I., Nienartowicz, K., Glass, F., Marchal, O., Grenon, M., Mowlavi, N., Holl, B., Clementini, G., Aerts, C., Mazeh, T., Evans, D. W., Szabados, L., Brown, A. G. A., Vallenari, A., Prusti, T., … Zwitter, T. (2019). Gaia Data Release 2. Variable stars in the colour-absolute magnitude diagram. *Astronomy & Astrophysics*, *623*, A110. https://doi.org/10.1051/0004-6361/201833304

Ivezić, Ž., Kahn, S. M., Tyson, J. A., Abel, B., Acosta, E., Allsman, R., Alonso, D., AlSayyad, Y., Anderson, S. F., Andrew, J., Angel, J. R. P., Angeli, G. Z., Ansari, R., Antilogus, P., Araujo, C., Armstrong, R., Arndt, K. T., Astier, P., Aubourg, É., … Zhan, H. (2019). LSST: From Science Drivers to Reference Design and Anticipated Data Products. *873*(2), 111. https://doi.org/10.3847/1538-4357/ab042c

Johnson, M. A. C., Gandhi, P., Chapman, A. P., Moreau, L., Charles, P. A., Clarkson, W. I., & Hill, A. B. (2019). Prospecting for periods with LSST - low-mass X-ray binaries as a test case. *Monthly Notices of the Royal Astronomical Society*, *484*(1), 19–30. https://doi.org/10.1093/mnras/sty3466

Kasliwal, V. P., Vogeley, M. S., & Richards, G. T. (2015). Are the variability properties of the Kepler AGN light curves consistent with a damped random walk? *Monthly Notices of the Royal Astronomical Society*, *451*(4), 4328–4345. https://doi.org/10.1093/mnras/stv1230

Kovačević, A. B. (2024). Two-dimensional (2D) hybrid method: Expanding 2D correlation spectroscopy (2D-COS) for time series analysis. *Applied Spectroscopy*, *0*(0), 00037028241241308. https://doi.org/10.1177/00037028241241308

Kovačević, A. B., Nina, A., Popović, L. Č., & Radovanović, M. (2022). Two-dimensional correlation analysis of periodicity in noisy series: Case of VLF signal amplitude variations in the time vicinity of an earthquake. *Mathematics*, *10*(22). https://doi.org/10.3390/math10224278

Kovačević, A. B., Pérez-Hernández, E., Popović, L. Č., Shapovalova, A. I., Kollatschny, W., & Ilić, D. (2018). Oscillatory patterns in the light curves of five long-term monitored type 1 active galactic nuclei. *475*(2), 2051–2066. https://doi.org/10.1093/mnras/stx3137

Kovačević, A. B., Popović, L. Č., & Ilić, D. (2020). Two-dimensional correlation analysis of periodicity in active galactic nuclei time series. *Open Astronomy*, *29*(1), 51–55. https://doi.org/10.1515/astro-2020-0007

Kovačević, A. B., Popović, L. Č., Simić, S., & Ilić, D. (2019). The Optical Variability of Supermassive Black Hole Binary Candidate PG 1302-102: Periodicity and Perturbation in the Light Curve. *871*(1), 32. https://doi.org/10.3847/1538-4357/aaf731

Kovačević, A. B., Radović, V., Ilić, D., Popović, L. Č., Assef, R. J., Sánchez-Sáez, P., Nikutta, R., Raiteri, C. M., Yoon, I., Homayouni, Y., Li, Y.-R., Caplar, N., Czerny, B., Panda, S., Ricci, C., Jankov, I., Landt, H., Wolf, C., Kovačević-Dojčinović, J., … Marčeta-Mandić, S. (2022). The LSST Era of Supermassive Black Hole Accretion Disk Reverberation Mapping. *262*(2), 49. https://doi.org/10.3847/1538-4365/ac88ce

Kovačević, A. B., Yi, T., Dai, X., Yang, X., Čvorović-Hajdinjak, I., & Popović, L. Č. (2020). Confirmed short periodic variability of subparsec supermassive binary black hole candidate Mrk 231. *494*(3), 4069–4076. https://doi.org/10.1093/mnras/staa737

DRAFT

Noda, I. (2018). Chapter 2 - advances in two-dimensional correlation spectroscopy (2DCOS). In J. Laane (Ed.), *Frontiers and advances in molecular spectroscopy* (pp. 47–75). Elsevier. https://doi.org/https://doi.org/10.1016/B978-0-12-811220-5.00002-2

Saxena, A., Salvato, M., Roster, W., Shirley, R., Buchner, J., Wolf, J., Kohl, C., Starck, H., Dwelly, T., Comparat, J., & al., et. (2024). CIRCLEZ : Reliable photometric redshifts for active galactic nuclei computed solely using photometry from Legacy Survey Imaging for DESI. *690*, A365. https://doi.org/10.1051/0004-6361/202450886

Vio, R., Cristiani, S., Lessi, O., & Salvadori, L. (1991). 3C 345: Is the Variability of Quasars Nonlinear? *380*, 351. https://doi.org/10.1086/170594

DRAFT